

Quantum Approximate Optimization Algorithm

E. Farhi, J. Goldstone, S. Gutmann, arXiv:1411.4028

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video

The Combinatorial Optimization Problem (COP) $\equiv$ QUBO

Quadratic Unconstrained Binary
Optimization

$$\mathcal{E}(z) = 2z_1z_2 + 2z_2z_3 + z_3z_1$$

$$2^3 = 8$$

$$\min_z \mathcal{E}(z), \text{ where}$$

Ising Energy
of
Spin Glass
system

$$\mathcal{E}(z) := \sum_{i,j=1}^n w_{ij} z_i z_j, = \sum w_{ij} (1-2x_i)(1-2x_j)$$

$$z := (z_1, \dots, z_n). \quad x = (x_1, \dots, x_n)$$

$$\mathcal{E}(z) = \sum_{i,j \in K} w_{ij} z_i z_j z_k$$

Weights: $w_{ij} \in \mathbb{R}$ define the problem.

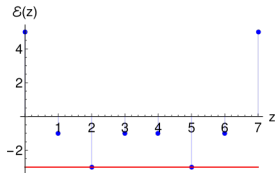
Decision variables: $z_i \in \{-1, 1\}$ define the solution space.

$$z = 1 - 2x$$

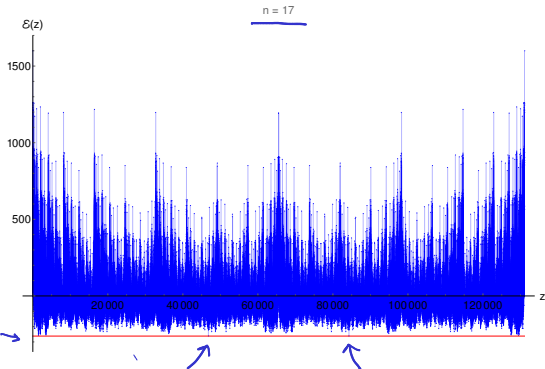
x	z
0	1
1	-1

Int	Bin	z	$\mathcal{E}(z)$
0	000	(1, 1, 1)	5
1	001	(1, 1, -1)	-1
2	010	(1, -1, 1)	-3
3	011	(1, -1, -1)	-1
4	100	(-1, 1, 1)	-1
5	101	(-1, 1, -1)	-3
6	110	(-1, -1, 1)	-1
7	111	(-1, -1, -1)	5

$$3 = n$$



video



① many local min

Exact is NP Hard

Number of variables $n = 17$

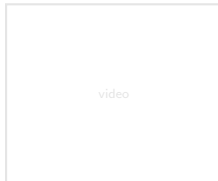
② Size of the solution space $2^n = 131072$

$$E_{\min} = -262$$

$$z_{\text{sol}} = \{46800, 84271\}$$

CPU time: 0.22 seconds

$$2^{200} = 1606938044258990275541962092341162602522202993782792835301376$$



Quantum formulation of the COP

- Replace n bits with n qubits and get the kets $|z_i\rangle$ corresponding to the variables $z_i \in \{-1, 1\}$.

$|z\rangle = |z_1\rangle \otimes \dots \otimes |z_n\rangle$, where $\{|z\rangle\}$ will form the computation basis of the n -qubit Hilbert space.

- Replace variable z_i in $\mathcal{E}$ with the Pauli operator Z_i and obtain the Hamiltonian:

$$H := \sum_{i,j=1}^n w_{ij} Z_i Z_j, \text{ where the operator } Z_i \text{ acts only on the } \underline{i\text{th}} \text{ qubit: } Z_i |z_i\rangle = z_i |z_i\rangle$$

$$\begin{aligned} Z|+\rangle &= +1|+\rangle \\ Z|-\rangle &= -1|-\rangle \end{aligned}$$

- Then, the energy expectation value will be

Diagonal elements

$$\langle z | H | z \rangle = \sum_{i,j=1}^n w_{ij} z_i z_j =: \underline{\mathcal{E}(z)}$$

$$Z_i Z_j = I \otimes \dots \otimes I \otimes Z \otimes I \otimes \dots \otimes I \otimes Z \otimes I \otimes \dots \otimes I$$

$\uparrow_{i\text{th}} \quad \quad \quad \uparrow_{j\text{th}}$

$$Z_i Z_j |z\rangle = z_i z_j |z\rangle$$

$$H |z\rangle = \mathcal{E}(z) |z\rangle$$

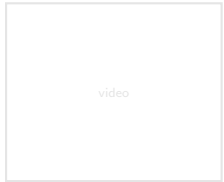
$\uparrow \quad \quad \uparrow$
Eigenvalue Eigenket

$O(n^2)$

- The exact diagonalization of H will not solve the problem **efficiently** as the size of H , 2^n , grows exponentially with the problem size n .

$$\underline{\underline{\delta(2^n)}}$$

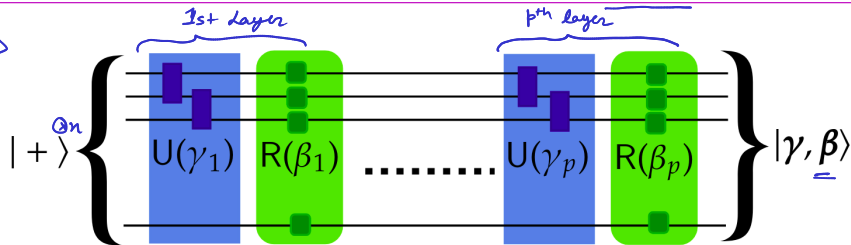
H is $2^n \times 2^n$ matrix



A. Lucas, Ising formulations of many NP problems, Front. Phys. 2, 5 (2014).

Quantum Approximate Optimization Algorithm (QAOA) for the COP

$$|+\rangle = \frac{|+1\rangle + |-1\rangle}{\sqrt{2}}$$



$$|\gamma, \beta\rangle = \prod_{k=1}^p R(\beta_k) U(\gamma_k) |+\rangle$$

$$U(\gamma) = \exp(-i\gamma \overset{\text{prob Hamiltonian}}{H}) = \prod_{i,j=1}^n \underbrace{\exp(-i\gamma w_{ij} Z_i Z_j)}_{U_{ij}(\gamma)}$$

$$R(\beta) = \exp(-i\beta S_x) = \prod_{i=1}^n \underbrace{\exp(-i\beta X_i)}_{R_i(\beta)}, \quad \text{where } S_x := \sum_{i=1}^n X_i \text{ is a mixing Hamiltonian.}$$

$$\mathcal{E}(\gamma, \beta) = \langle \gamma, \beta | H | \gamma, \beta \rangle$$

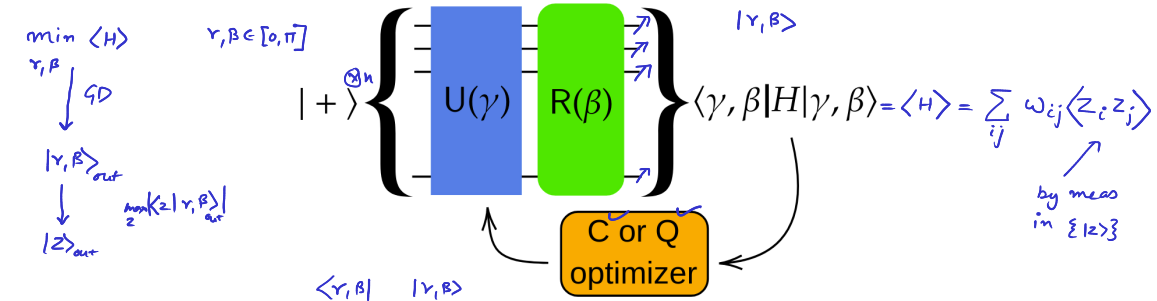
$$\gamma = (\gamma_1, \gamma_2, \dots, \gamma_p)$$

$$\beta = (\beta_1, \beta_2, \dots, \beta_p)$$

$$\exp(A+B) = \exp(A) \exp(B) \quad \text{if } AB=BA$$

video

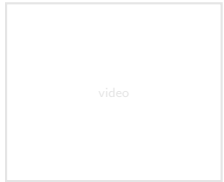
$\underline{U(r+\pi)} = \overline{\uparrow} \underline{U(r)}$



$$\langle H \rangle = \langle \gamma, \beta | H | \gamma, \beta \rangle = \langle + | U^\dagger R^\dagger H R U | + \rangle$$

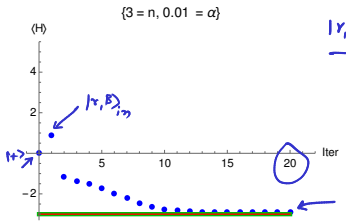
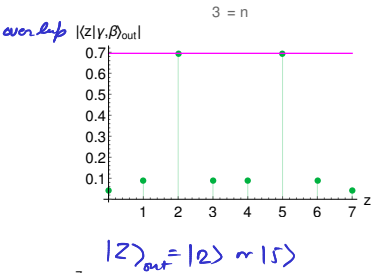
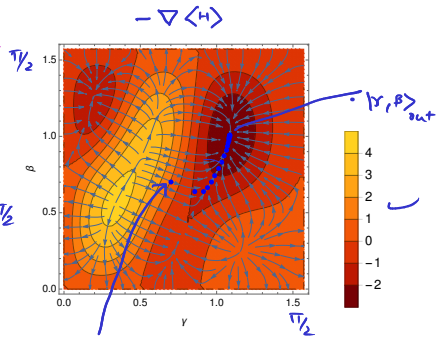
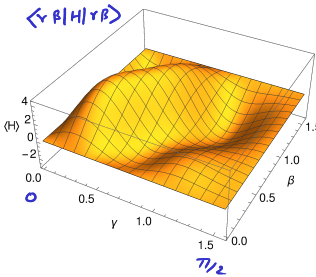
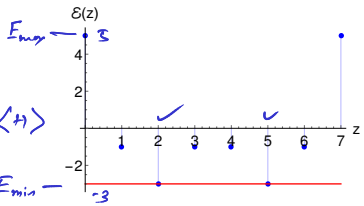
$$\rightarrow \frac{\partial \langle H \rangle}{\partial \gamma} = \langle + | \frac{\partial U^\dagger}{\partial \gamma} R^\dagger H R U | + \rangle + \langle + | U^\dagger R^\dagger H R \frac{\partial U}{\partial \gamma} | + \rangle$$

$$\rightarrow \frac{\partial \langle H \rangle}{\partial \beta} = \langle + | U^\dagger \frac{\partial R^\dagger}{\partial \beta} H R U | + \rangle + \langle + | U^\dagger R^\dagger H \frac{\partial R}{\partial \beta} U | + \rangle$$

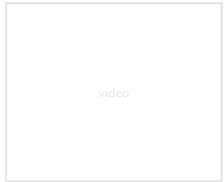


π_{AI} Single ($p = 1$) layer QAOA for $n = 3$ qubits

$2^3 = 8$
 $E(z) = 2Z_1Z_2 + 2Z_2Z_3 + Z_3Z_1$
 $H = 2Z_1Z_2 + 2Z_2Z_3 + Z_3Z_1$
 $3 = n$



$(\gamma, \beta)_i = (0.7, 0.7)$
 $|\gamma, \beta\rangle_{out} = \sum_z |z\rangle \langle z | \gamma, \beta \rangle_{out}$
 Probomp



$Q\pi_{AI}$ Computing gradient on a quantum computer by the parameter sift rule

$AB = BA$ Commut.

$$\frac{\partial \langle H \rangle}{\partial \gamma} = \langle + | \sum_{i,j} \frac{\partial U_{ij}^\dagger}{\partial \gamma} \prod_{(i',j') \neq (i,i)} U_{i'j'}^\dagger R^\dagger H R U | + \rangle + \langle + | U^\dagger R^\dagger H R \sum_{i,j} \frac{\partial U_{ij}}{\partial \gamma} \prod_{i',j'} U_{i'j'} | + \rangle$$

because Inner prod is linear

$$h(r) = f(r)g(r)$$

$$h' = f'g + g'f$$

$$U = \prod_{ij} U_{ij} \quad i = \sqrt{-1}$$

$$= \sum_{i,j} \left[\langle + | \frac{\partial U_{ij}^\dagger}{\partial \gamma} \prod_{i',j'} U_{i'j'}^\dagger R^\dagger H R U | + \rangle + \langle + | U^\dagger R^\dagger H R \prod_{i',j'} U_{i'j'} \frac{\partial U_{ij}}{\partial \gamma} | + \rangle \right]$$

$$U = \exp(-it\phi)$$

$$\frac{\partial U}{\partial t} = -i\phi \exp(-it\phi)$$

Schrodinger Eq

parameter sift rule

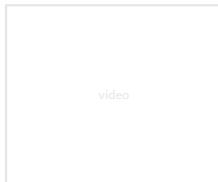
$$= \sum_{i,j} \left[\langle + | \frac{\partial U_{ij}^\dagger}{\partial \gamma} H_{ij} U_{ij} | + \rangle + \langle + | U_{ij}^\dagger H_{ij} \frac{\partial U_{ij}}{\partial \gamma} | + \rangle \right] = i \sum_{ij} \omega_{ij} \langle + | U_{ij}^\dagger [\tilde{z}_i \tilde{z}_j H_{ij} - H_{ij} z_i z_j] U_{ij} | + \rangle$$

$i\omega_{ij} z_i z_j U_{ij}$ $-i\omega_{ij} z_i z_j U_{ij}$

$$= \sum_{i,j} w_{ij} \left[\langle + | U_{ij}^\dagger \left(\gamma + \frac{\pi}{4w_{ij}} \right) H_{ij} U_{ij} \left(\gamma + \frac{\pi}{4w_{ij}} \right) | + \rangle - \langle + | U_{ij}^\dagger \left(\gamma - \frac{\pi}{4w_{ij}} \right) H_{ij} U_{ij} \left(\gamma - \frac{\pi}{4w_{ij}} \right) | + \rangle \right]$$

$$i(\sigma H - H \sigma) = \underbrace{U_{ij}^\dagger \left(\frac{\pi}{4w_{ij}} \right)}_{\left(\frac{I+i\sigma}{\sqrt{2}} \right)} H \underbrace{U \left(\frac{\pi}{4w_{ij}} \right)}_{\frac{I-i\sigma}{\sqrt{2}}} - \underbrace{U_{ij}^\dagger \left(-\frac{\pi}{4w_{ij}} \right)}_{\frac{I-i\sigma}{\sqrt{2}}} H \underbrace{U_{ij} \left(-\frac{\pi}{4w_{ij}} \right)}_{\frac{I+i\sigma}{\sqrt{2}}}$$

$U_{ij} = \exp(-i\sigma U_{ij} z_i z_j)$



K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii, Phys. Rev. A **98**, 032309 (2018).

M. Schuld, V. Bergholm, C. Gogolin, J. Izaac, and N. Killoran, Phys. Rev. A **99**, 032331 (2019).

Traveling Salesman Problem (TSP)

$$\left(\sum_{t=1}^N z_{t,u} + (N-2) \right)^2 = 0^2 = 0$$



Total number of nodes (cities) to travel $N = 15$

Cost (distance) of traveling from node u to v : $c_{uv} \geq 0$

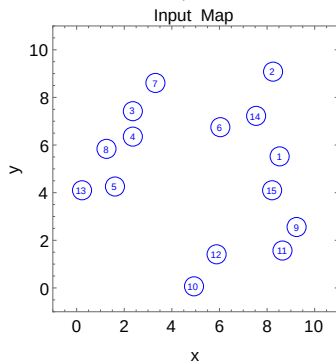
time steps $t = 1, \dots, N + 1$

Decision variable $z_{t,u} = \begin{cases} +1 & \text{if at time } t \text{ salesman is at node } u \\ -1 & \text{otherwise} \end{cases}$

time $\rightarrow t$ *place* $\rightarrow u$

Total cost $C = \sum_{u,v=1}^N c_{uv} \sum_{t=1}^N \left(\frac{1+z_{t,u}}{2} \right) \left(\frac{1+z_{t+1,v}}{2} \right)$

1 or 0 *1 or 0*



Combinatorial optimization: $\min C$ such that

$\rightarrow \sum_{t=1}^N z_{t,u} = -1$ *+1 -1 -1 -1 ... -1 = 1 - (N-1)* every node u , must be visited only once

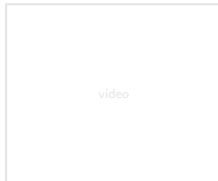
$\sum_{u=1}^N z_{t,u} = -1$ at each t , salesman must be at one node



number of var = $N^2 = 15^2$

Size of sol-space: $2^{(15^2)} =$

53919893334301279589334030174039261347274288845081144962207220498432





Quantum (QAOA) formulation of the TSP

$z_{t,u}$

- Take $n = N^2$, replace n bits with n qubits, and get the ket $|z\rangle$ corresponding to every decision variable $z \in \{-1, 1\}$.

$$H = a H_a + b H_b$$

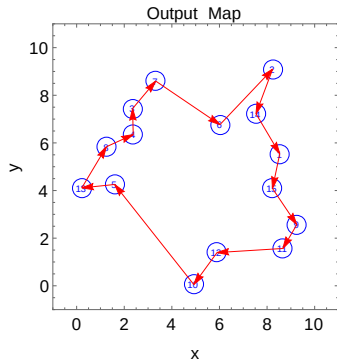
$$H_a := \underbrace{\sum_{u,v=1}^N c_{uv} \sum_{t=1}^N \left(\frac{1 + Z_{t,u}}{2} \right) \left(\frac{1 + Z_{t+1,v}}{2} \right)}_{\text{Cost}}$$

$$H_b := \underbrace{\left[\sum_{u=1}^N \left(\sum_{t=1}^N Z_{t,u} + (N-2) \right)^2 + \sum_{t=1}^N \left(\sum_{u=1}^N Z_{t,u} + (N-2) \right)^2 \right]}_{\text{Constraints}}$$

$|z\rangle = |z_1\rangle \otimes \dots \otimes |z_n\rangle$ are the energy eigenkets of H .

- $S_x := \sum_{i=1}^n X_i$ is a mixing Hamiltonian, and $|+\rangle = \frac{1}{\sqrt{2^n}} \sum_z |z\rangle$ is its eigenket.

a, b defines the imp of H_a, H_b



SA



$$3 \times 3 = N^2 = 9$$

$$4 \times 4 = N^2 = 16$$

